

Automatic Vehicle Number Plate Detection and Recognition System using YOLOv8n and EasyOCR

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Abstract—In real-time applications such as heavy traffic surveillance and gate pass automation in different mining industries, vehicle number plate recognition is a crucial task due to an increase in traffic, dirt, and bad weather. So manual tracking of all the vehicle number plates at the entry and exit gates in mining industries as well as in different traffic surveillance areas is not possible without a proper number plate detection and recognition model. So, to avoid such scenario, a smart traffic system is needed to recognize different vehicle number plates efficiently. Due to the presence of complexity in Indian heavy vehicle number plates, we have proposed an Automatic Vehicle Number Plate Recognition (AVNPR) model. We have trained and validated our proposed model in a single dataset of 20293 different vehicle images. As the YOLOv8n algorithm has not been tested in real-time, so we have used YOLOv8n for the detection purpose and got a mean Average Precision (mAP) of 98.8%. Further, the recognition is done by using the EasyOCR engine with an average recognition accuracy of 92.32%. Our proposed AVNPR model is found to be better than that of the other existing models in terms of average detection and recognition accuracy.

Keywords—Vehicle Number Plate, Detection, Recognition, EasyOCR, YOLOv8n, Mining Industry, Convolutional Neural Network

I. INTRODUCTION

Now-a-days the population size, connectivity of roads, and the number of vehicles is increasing day-by-day in developing countries like India. Due to which the efficiency of the traffic system may decrease. Also, the traffic rules are not followed by all the people in different countries due to the lack of smartness in traffic systems. The traffic systems in other developing countries are very strict due to the usage of different automated number plate recognition systems. The number of accidents may also be reduced using of different number plate recognition systems. So, to overcome the traffic related problems in real time such as frequent accidents, stolen vehicle identification, and identification of vehicle owners, we need to detect and recognize the vehicle number plate in an efficient manner. Hence, different Deep Learning models are proposed by different authors [1] to improve the accuracy and efficiency of the traffic system significantly. The challenges

related to number plate detection and recognition need further improvement for different real time applications. In this regard, many vehicle number plate detection and recognition models are proposed [2] like Single Shot Detection (SSD), You Only Look Once (YOLO), CNN, and Faster R-CNN. Out of all the models, YOLO [3] is very suitable for number plate detection in different real time applications.

In this regard, Islam et al. [4] have proposed two Deep Convolutional Neural Network (DCNN) models to detect and identify the vehicle license plates from real-time videos. The first CNN model detected the number plate from real-time video and the second CNN model detected the characters on the license plate. They have trained their model on BanglaLekha-Isolated handwritten character dataset. Their model is only fit for the Bangla vehicle license plates and not applicable for other language number plates. They have also not used YOLO model and OpenCV library for the detection of number plates. Further, the same problem is addressed by Shreya et al. [5] and they have proposed an Automatic Number Plate Recognition (ANPR) system by using YOLOv5 object detection algorithm to detect the number plates containing alphabets and digits. They have used a database of 250 images to train the YOLOv5 algorithm. Still, they need a good dataset annotation for the validation of their model. The efficiency and accuracy of their model needs to improve further.

Further, Atikuzzaman et al. [6] have proposed a number plate detection and recognition model, which is composed of three phases like number plate detection, class letter segmentation and recognition. These three phases adopted HAAR feature based classifier, class letter extractor and CNN for detection and recognition. They have taken 5500 license plates in their data set. Due to complexity, they have not used any RGB images. And, they have implemented their model only in Bengali ANPR systems in bad weather and low light conditions. To avoid such restrictions, Recently, Soundarya et al. [7] have proposed an ANPR model to recognize alpha numeric vehicle number plate with the help of Deep Learning algorithm. A coherent model, Faster R-CNN is used in their proposed model to recognize the number plate, and the alpha numeric digits are then extracted from the number plates using an Optical Character Recognition (OCR) method. They have

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not used any YOLO model in their proposed work.

To address the same problem, Vaidehi et al. [8] have proposed a vehicle number plate detection method to recognize a suspicious vehicle in a particular area. They have trained the image database on YOLOv4 model in different light illumination conditions. Further, the OCR engine helps to recognize each character of the vehicle number plate of different suspicious vehicles. Subsequently, a novel security surveillance system [9] for no parking detection using vehicle number plate recognition has been proposed for metropolitan area. They have used a Region based CNN (R-CNN) algorithm for vehicle identification. The OCR system is used for number plate character recognition in different real-time environments. There is a scope to improve the model accuracy as they have not considered the concept of faster R-CNN. Therefore, another number plate detection method has been proposed by Palanivel et al. [10] in which different surveyance cameras are placed in the traffic areas. They have used Faster R-CNN algorithm for detection purposes. Using frame segmentation and image interpolation techniques, the number plates are detected. Thereafter, the number is extracted from the image by OCR technique.

To improve the accuracy of the number plate detection model, Shrivastava et al. [11] have proposed a Vehicle Recognition Number Plate Recognition (VRNPR) system to detect and recognize the number plates using CNN strategy. The OCR method is used for character recognition of the number plates. But the authors have not used the YOLO model and OpenCV library for the detection of vehicle registration number plates. In this regard, Shasidhar et al. [12] have proposed a vehicle number plate detection and recognition model implemented using YOLOv3 model for Region of Interest (RoI). They have captured blurred images of different kind of vehicles and applied them to different machine learning models for further processing. Their model is validated by fetching the data from Indian RTO database and got different accuracy percentage value when compared to other number plate detection models. They have not captured any real-time videos to detect the number plate of any vehicle from those video frames.

In different mines like iron ore mines, coal mines, manganese ore mines, and limestone mines, different types of heavy vehicles are engaged to carry the material from the mines to the destination. Due to heavy dirt and dust in mine areas, the number plates of those vehicles are very difficult to detect and recognize by the traffic system as well as by the RFID reader. Due to some technical problem, the RFID reader may fail to read some RFID tags attached to mining heavy vehicles. In most cases, heavy vehicles are responsible for making various types of accidents by violating traffic rules. Hence the detection and recognition of such vehicle number plates is necessary to avoid such scenarios. Therefore, to detect and recognize such complex vehicle number plates, we have proposed a vehicle number plate detection and recognition model to enhance the automation of entry and exit gates, the traffic surveillance and traffic security in different real time applications. Recently, the YOLOv8 algorithm is used by [13] for the enhancement of traffic monitoring system but the YOLOv8 algorithm has not been used for the vehicle number plate detection in real-time. In this regard, we have proposed an Automatic Vehicle Number Plate Recognition

(AVNPR) model, in which the vehicle number plates are detected using YOLOv8n algorithm [14] and the recognition of the number plates are done by an EasyOCR engine, which helps to detect the alpha-numeric characters of the number plates. The proposed algorithm is implemented using Python programming and Visual Studio Code editor. By implementing both the notions of YOLOv8n algorithm and EasyOCR engine for real-time mining heavy vehicles, the performance of the proposed model in terms of detection and recognition accuracy has been found to be superior to that of the other existing models.

The remaining of this paper is organized as follows. Section II describes the proposed AVNPR model and presents the model detection and recognition using YOLOv8n algorithm and EasyOCR engine. The results and discussion details are presented in Section III. Finally, the conclusion and future works are included in Section IV.

II. PROPOSED AUTOMATIC VEHICLE NUMBER PLATE RECOGNITION (AVNPR) MODEL

An Automatic Vehicle Number Plate Recognition (AVNPR) model has been proposed for the detection and recognition of different types of number plates in real-time environment using YOLOv8n algorithm and EasyOCR engine. Thus, the number plate detection is done by the implementation of YOLOv8n algorithm and the recognition is done by using EasyOCR engine. The block diagram of the proposed AVNPR model is shown in Fig. 1. As shown in Fig. 1, we have taken 21175 input images of different vehicles in both day and night scenarios. Initially, we captured the images of real-time mining heavy vehicles with a bullet camera and then the corresponding number plates are detected by YOLOv8n algorithm. Then the enhancement of the number plate images is done by using OpenCV to get the required information of any vehicle number plate. The details of image enhancement procedure are explained in Fig. 6 and 7. We have taken a video of foreign vehicles from Roboflow website [15] and captured some real-time mining truck images with a bullet camera installed on a mounting pole from a mining industry for implementation purpose. Further, the recognition of the preprocessed enhanced number plate images is done by using EasyOCR engine.

A. Detection of Vehicle Number Plates using YOLOv8n Algorithm

The raw image data is processed and annotated by Roboflow [15]. We have taken 21175 different annotated vehicle images to create single dataset for training, validation, and testing of the model. After training, validation, and testing of the model, we have given a video of foreign vehicles as input with different illumination conditions with different view angles. The Indian number plates are more complex than the foreign vehicle number plates, therefore the Indian number plates are difficult to detect in real-time environment. Hence, we have studied different complex number plates by capturing some heavy vehicle images inside the mining industry by the help of bullet cameras attached to different mounting poles. YOLOv8 algorithm is the recent version of YOLO with the addition of new features to enhance efficiency, performance and flexibility. YOLOv8 is added with a full range of vision

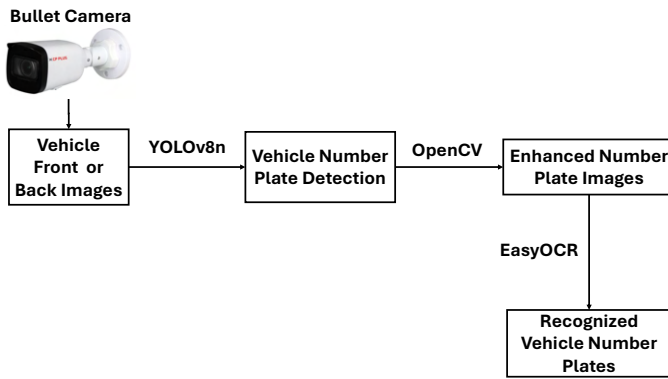


Fig. 1: Proposed AVNPR model for vehicle number plate detection and recognition.

AI tasks such as segmentation, detection, classification and tracking for different domains and applications. From the literature, we can know that the YOLOv8n algorithm has not been used for mining vehicle number plate detection purposes. So, to explore the capabilities, performance, and the efficiency of the YOLOv8 algorithm, we have used the algorithm for our proposed model for the detection of the mining vehicle number plates for different real-time applications in different environmental scenarios.

B. Enhancement of Number Plates using OpenCV

The vehicle number plate is detected by the AVNPR model using YOLOv8n algorithm as shown in Fig. 6 and 7 and then the image enhancement is done through a preprocessing method using OpenCV. In the initial step of preprocessing method, the cropped RGB image is converted into a grayscale image which simplifies the image data by reducing the number of three-color channel to one-color channel. This makes the next processes more efficient in nature. Further, the grayscale image is converted into a binary image using an adaptive threshold method which turns the image into black and white format, where the text and other features are highlighted against the background. The binary image is then analyzed to detect the lines and calculate the median angle of rotation and then the binary image is deskewed to correct any skew present in the image and converted into a deskewed RGB image. Again, similar process is done for the next steps i.e. the deskewed RGB image is converted into deskewed gray image and then the deskewed gray image is converted into deskewed binary image which processed further and finally given to the EasyOCR engine for the recognition of the vehicle number plates.

C. Recognition of Number Plates using EasyOCR Engine

EasyOCR is a type of Optical Character Recognition (OCR) engine which helps to extract the text from a detected image of a vehicle number plate. The deskewed binary image as shown in Fig. 6 and 7 is the input image for the EasyOCR engine, in which each alphabet and each letters are extracted from the deskewed binary image to get the corresponding vehicle number plate. Hence, in this way different vehicle

number plates are recognized by the EasyOCR engine in an efficient way with high accuracy.



Fig. 2: Input image frames of foreign vehicles from a video at different angles.



Fig. 3: Recognized output image frames of foreign vehicle number plates.



Fig. 4: Input images of Indian heavy vehicles with different illumination inside a mining industry.



Fig. 5: Output images of Indian heavy vehicles with recognized number plates.

III. RESULTS AND DISCUSSION

For the validation of the efficiency of the proposed AVNPR model, the input foreign vehicle image frames as shown in Fig. 2 and mining Indian heavy vehicles as shown in Fig. 4 have been considered for the number plate detection purpose. The proposed AVNPR model is trained in Amazon Web Service (AWS) cloud with instance type g4dn.xlarge with one GPU and four VCPU using YOLOv8n algorithm in Ubuntu server and Ubuntu environment. Other system tool details have been shown in Table. I. The proposed AVNPR model takes care of the detection and recognition of different Indian as well as foreign vehicle number plates in daytime as well as in nighttime to enhance the security feature in different real time industrial applications. We have considered total 20293 images for training and validation purpose, and we have shown six foreign vehicle image frames as shown in Fig. 2 and the corresponding recognized output number plate images of corresponding foreign vehicles are shown in Fig. 3. Similarly, we have also taken six Indian heavy vehicle number plates in different illumination scenario inside a mining industry as shown in Fig. 4 and the recognized output number plates are shown in Fig. 5. The truck number plates are Indian number plates and complex in nature which consists of nine or ten alpha-numeric numbers and some extra characters along with the number plate. This complexity present in the Indian number plates has a direct impact on the detection and recognition accuracy of our proposed AVNPR model. We have used

TABLE I: Online System Platform Details

System Tools	Configuration
Cloud Platform	Amazon Web Server
GPU Architecture	Nvidia T4 Tensor Core
Operating System	Ubuntu
Hard Disk Storage	512 GB SSD
RAM	16GB
No of VCPU	4
No of GPU	1
GPU Library	CUDA 12.1, CUDNN 8.9.7
Framework	Ultralytics(YOLOv8n)

a database of 21175 different types of vehicle images and trained, validated, and tested them using YOLOv8n algorithm in cloud server as shown in Table. I. The size of input image is taken as 640×640 . The proposed AVNPR model is trained using one class only i.e. vehicle number plate. The configuration parameters of YOLOv8n are as shown in Table. II and the other parameter values are set as default. The mean Average Precision (mAP) as shown in Tab. V is calculated after 80th epoch and the highest mAP YOLO weight is selected.

A. Accuracy Analysis of AVNPR Model

The vehicle number plate is detected using YOLOv8n algorithm and got a detection accuracy of 98.8% mAP (Mean Average Precision) with an average inference time of 1.9ms which resulted a speed of 526.32 frames per second as shown in Table V i.e the proposed AVNPR model can predict 98.8% of correct prediction as shown in Fig. 8. Similarly, the precision value is also shown in Fig. 9. Besides that, the average loss

TABLE II: Model Training Parameters for YOLOv8n

Parameters	Value
Batch size	16
Initial Learning Rate	0.01
Max batch	2000
Optimizer	Stochastic Gradient Descent (SGD)
Momentum	0.937
Steps	Default

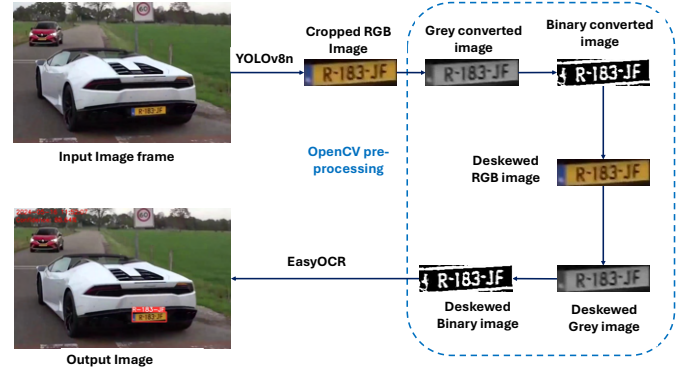


Fig. 6: Various stages of proposed AVNPR model for foreign vehicle number plates.

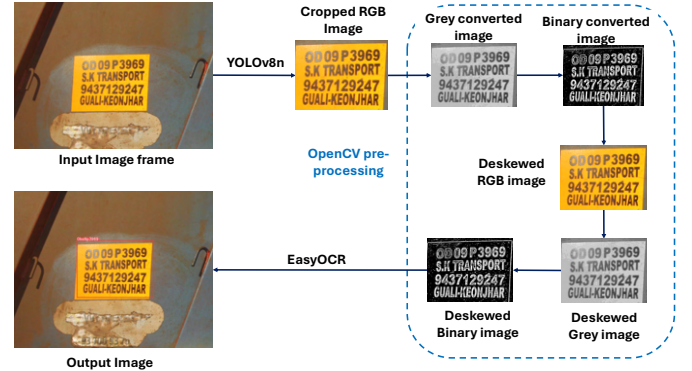


Fig. 7: Various stages of proposed AVNPR model for Indian mining heavy vehicle number plates.

is about 0.433% during the training of 80 epochs as shown in Fig. 10. The vehicle number plate recognition has been done using EasyOCR engine. The average recognition accuracy is calculated using (1) i.e the number of recognized characters by the total number of input characters. The average recognition accuracy for different foreign and Indian vehicles are shown in Table IV. Thus, the average recognition accuracy is 92.32% using EasyOCR engine for the recognition of different Indian and foreign vehicle number plates as shown in Table V. The EasyOCR engine has a good learning ability. The detection and recognition accuracy for different models are shown in Table III and Table V for comparison purposes. The performance

of our proposed model is better in terms of detection and recognition accuracy for complex number plates inside the mining industry.

$$\text{Recognition Accuracy} = \frac{\text{Recognized Characters}}{\text{Total Input Characters}} \quad (1)$$

TABLE III: Accuracy comparison of proposed AVNPR model with other models

Reference no.	Approach	Detection Accuracy(%)	Recognition Accuracy(%)
[16]	Morphological Operation and CNN	93.78	97.03
[17]	YOLO	95	92.70
[18]	Connected Component Analysis	95	NA
[19]	Neural Network	84	80
[6]	HAAR-Feature and CNN	96.92	91.38
Proposed AVNPR model	YOLOv8n	98.8	92.32

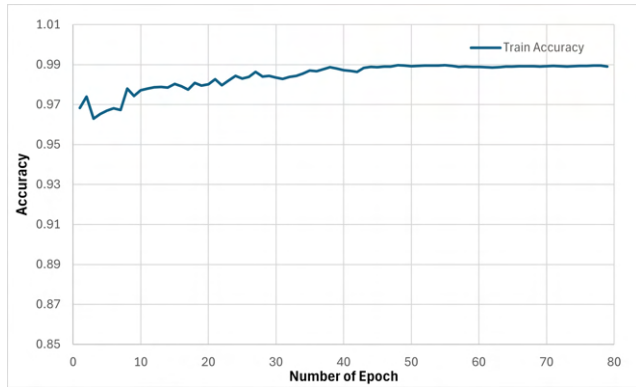


Fig. 8: Train accuracy for AVNPR model.

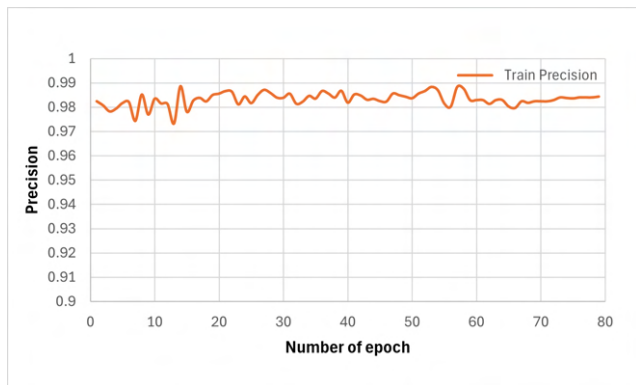


Fig. 9: Precision for AVNPR model.

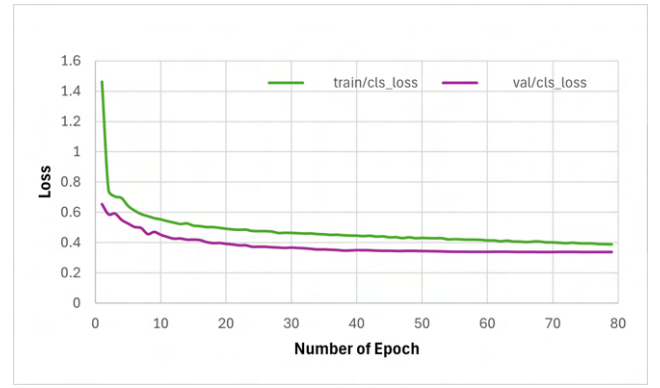


Fig. 10: Loss for AVNPR model.

TABLE IV: Efficiency of recognized number plate images

Input Numberplate	Recognized Numberplate	Recognition Accuracy (%)
R-183-JF	R-183-JF	100
N-894-JV	N894-JV	100
L-656-XH	L656XH	100
H-644-LX	H644LX	100
K-884-RS	K884-RS	100
66-HH-07	66-HH:07	100
OD09P 3969	Oba9p 3969	77.78
OD33AM 6762	0033AO 6762	80
OD33AM 6705	206337AN 6705	80
OD02CF 0963	ON02CF U963	80
OD33AM 6730	20D33AN 6730	90
OD09P 2002	OD09P 2002	100

TABLE V: Comparison of proposed AVNPR model with existing models

Detection models	mAP (%)	Recognition method	Accuracy (%)
SSD	74.3	Tesseract OCR	94.36
Faster RCNN	99.1	Template Matching	NA
YOLO v3	92	ResNet20 based CNN	85
YOLO v3	96	Custom 9 layer of CNN	91.05
YOLOv8n	98.8	EasyOCR	92.32

IV. CONCLUSION

In this paper, we have proposed a AVNPR model for different foreign vehicles and Indian mining heavy vehicles for the detection and recognition of different complex number plates. The detection of the number plates is done using YOLOv8n algorithm, which is based upon real time Deep learning algorithm. We have taken 21175 images in one database of different heavy vehicles, four-wheeler and two-wheeler in various illumination conditions for training, validation, and testing purposes. Our proposed model has been tested for both daytime and nighttime images of different vehicle number plates. The accuracy of the proposed model for vehicle number plate detection is approx. 98.8%. But the recognition accuracy is 92.32% for the proposed model for

different number plates. We have chosen EasyOCR engine for the character recognition of different number plates. This engine is very efficient to give better results as compared to the other techniques. The performance of the proposed AVNPR model is better as compared to other existing models as shown in Table III in terms of detection and recognition accuracy of different number plates in different complex environment for different mining industry applications. The proposed model is robust in nature for the detection and recognition of any complex number plates in terms of font, size, color, and contrast with good accuracy. In future, our work will be extended to a vehicle gate pass automation system in different mining industry applications. This model will work at the time of failure of RFID reader at mining entry gates or at different weighbridge operations. Also to enhance traffic security, we will combine the AVNPR model with the concept of vehicle make and model recognition. In future we will enhance the accuracy of this model by adopting different postprocessing concepts.

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